

Day 3: A Server with Node.js and Express

Goal: Students build the Notes Board server with Express. The notes live on the server and the page reads, adds and deletes them with `fetch`. Reading is public. Adding and deleting need a password that is kept in a private `.env` file and never committed to Git.

Folder	Files
notes/	Day-3-Theory (read before class)
lab/	Day-3-Lab-Sheet (10 parts, plus Part 0)
help/	Day-3-Stuck-Sheet
starter/	Everything finished up to Day 2 (<code>notes-board</code> , <code>hello-server</code> , <code>setup-check.js</code>)
done/	<code>notes-board</code> with <code>server.js</code> , <code>data.js</code> , <code>package.json</code> , <code>.env.example</code> , <code>.gitignore</code> and the <code>public</code> folder

Highest risk of the day

`npm install express` on 30 to 60 computers at the same time over the lab network. Do Pilot B in `../INSTRUCTOR-PILOT-CHECKLIST.md` before this class. The instructor sheet for the day (with the fallback plan) is in `../00-Instructor-Kit/instructor-sheets/Day-3-Run-Sheet.md`.

What to know about the `done` folder

- In the Git repository it has **no** `node_modules` folder and **no** `.env` file, because Git ignores both. (A copy on the instructor's computer may still hold a `node_modules` folder left from testing. Give students a clean copy without it, so that they practise `npm install`.) Students run `npm install` and copy `.env.example` to `.env` (the Stuck Sheet has the steps).
- At the end of Day 3 the notes are kept in the memory of the server. They are lost when the server stops. This is on purpose: it is the reason for the database on Day 4.

Homework for Day 4

- Students check that their GitHub account exists and write down the username.
- Students run their Notes Board once more and add a note with their password.
- Students answer the "Check yourself" questions in the Day 3 Theory.
- Students try the extra challenges.
- Students write down at least two questions.